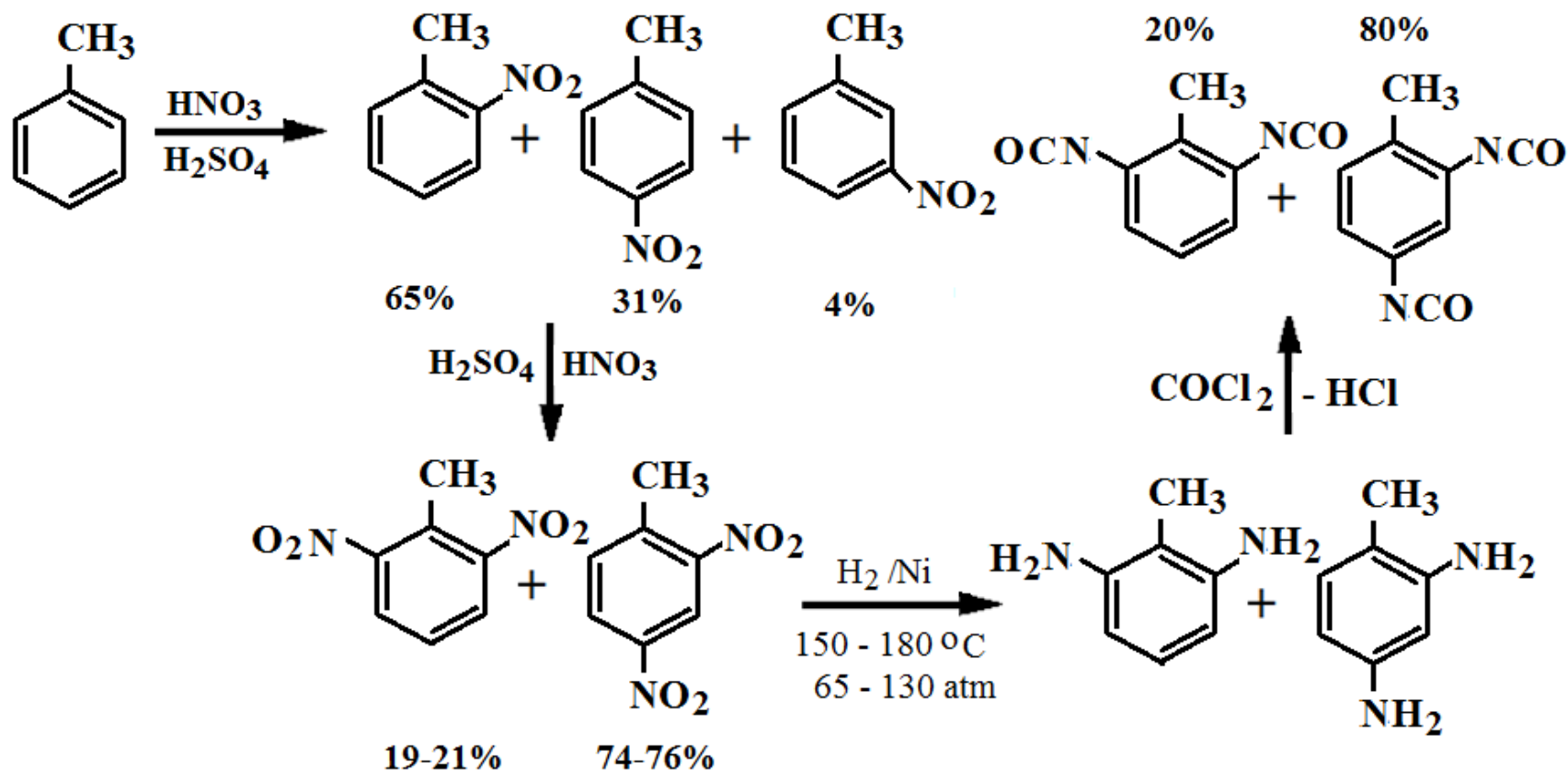


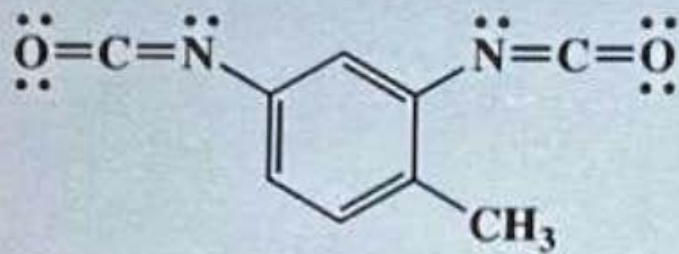
# Petrochemical Technology

## **Chemicals from Toluene**

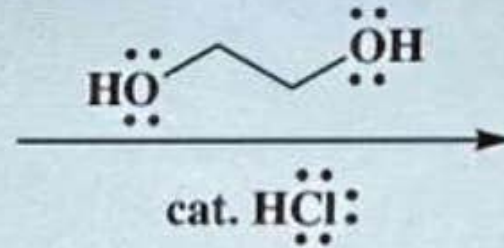
# Toluene Diisocyanate (TDI) from Toluene



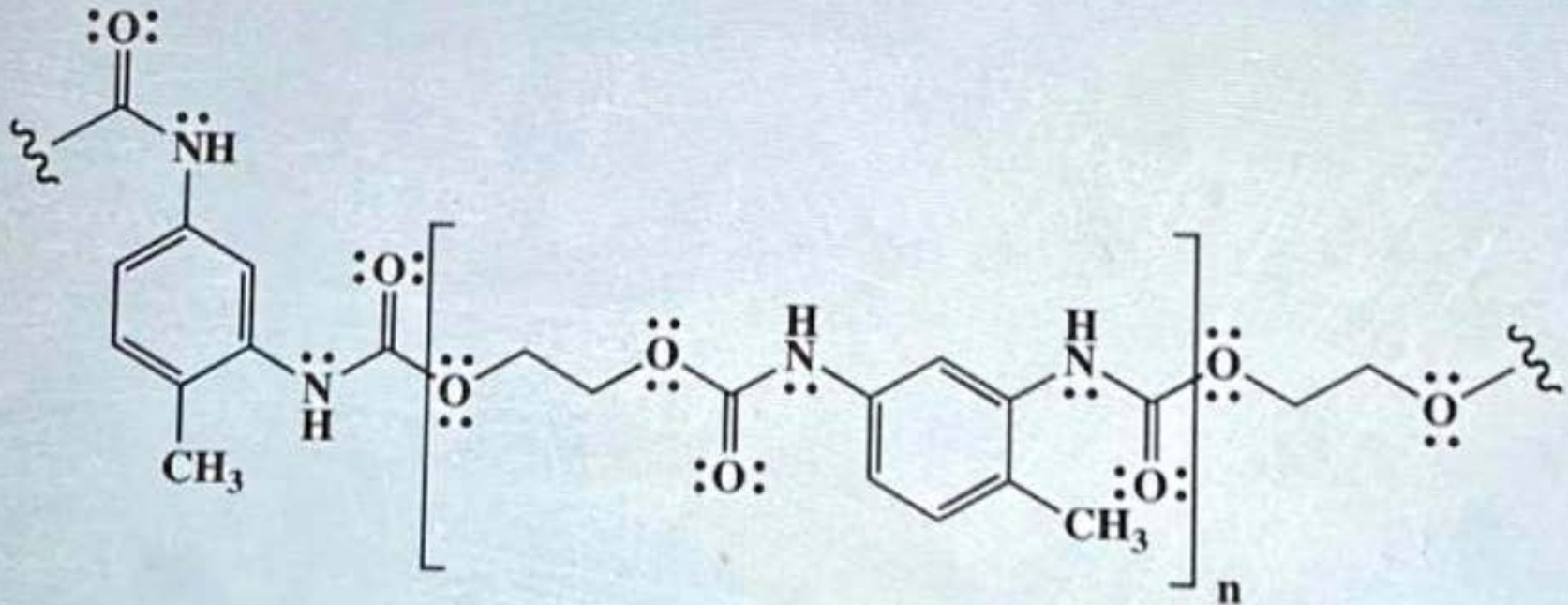
Jeeru, Lakshmana Rao; Saikumar Reddy R., V.; Pradhan, Sayantan; Kundu, Gautam; and Pradhan, Narayan C. Kinetics of Solid Acids Catalyzed Nitration of Toluene: Change in Selectivity by Triphase (Liquid-Liquid-Solid) Catalysis. *Asia-Pacific J. Chem. Engg.* 2018, 6 (6), 870-878.



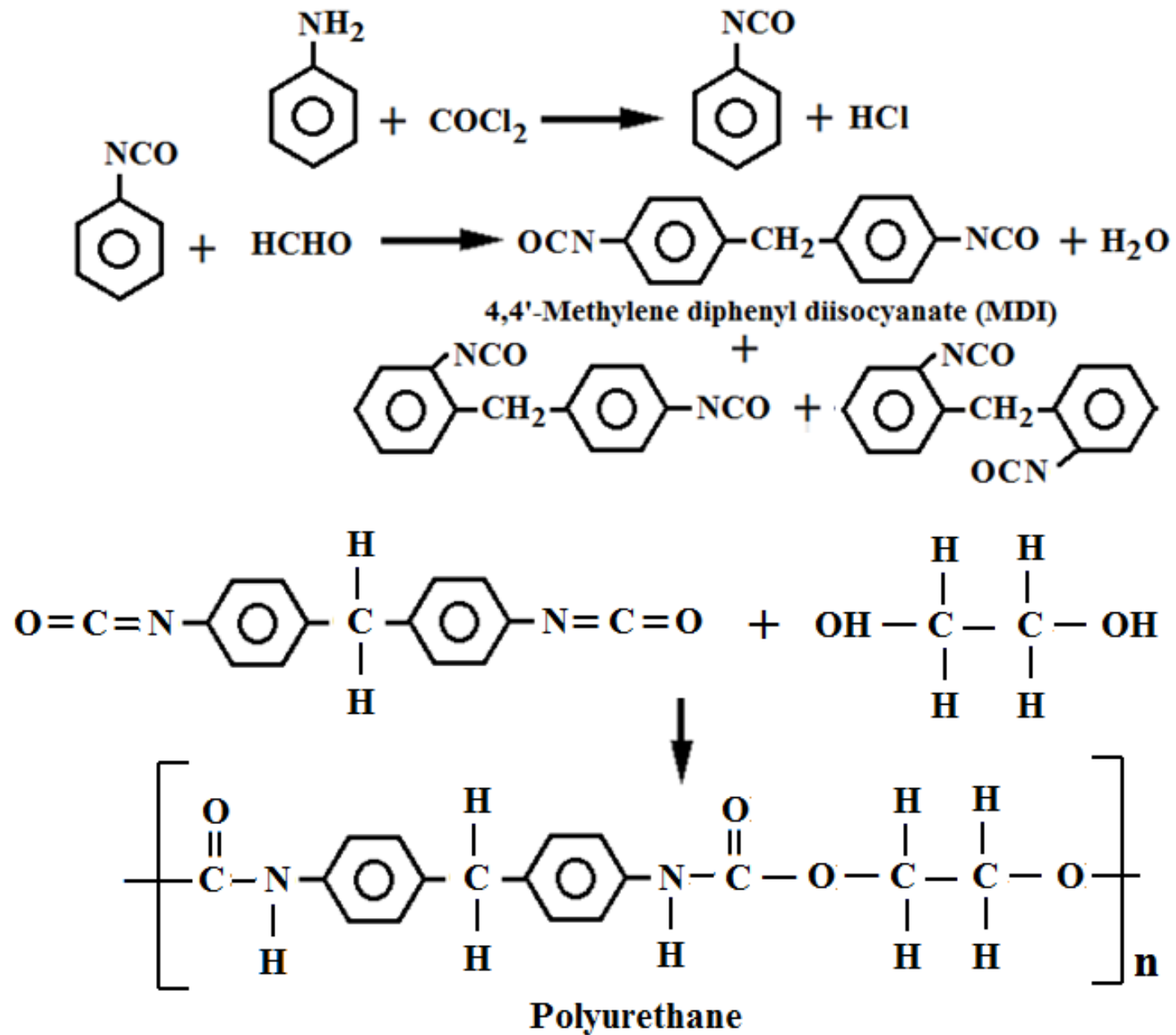
2,4-TDI



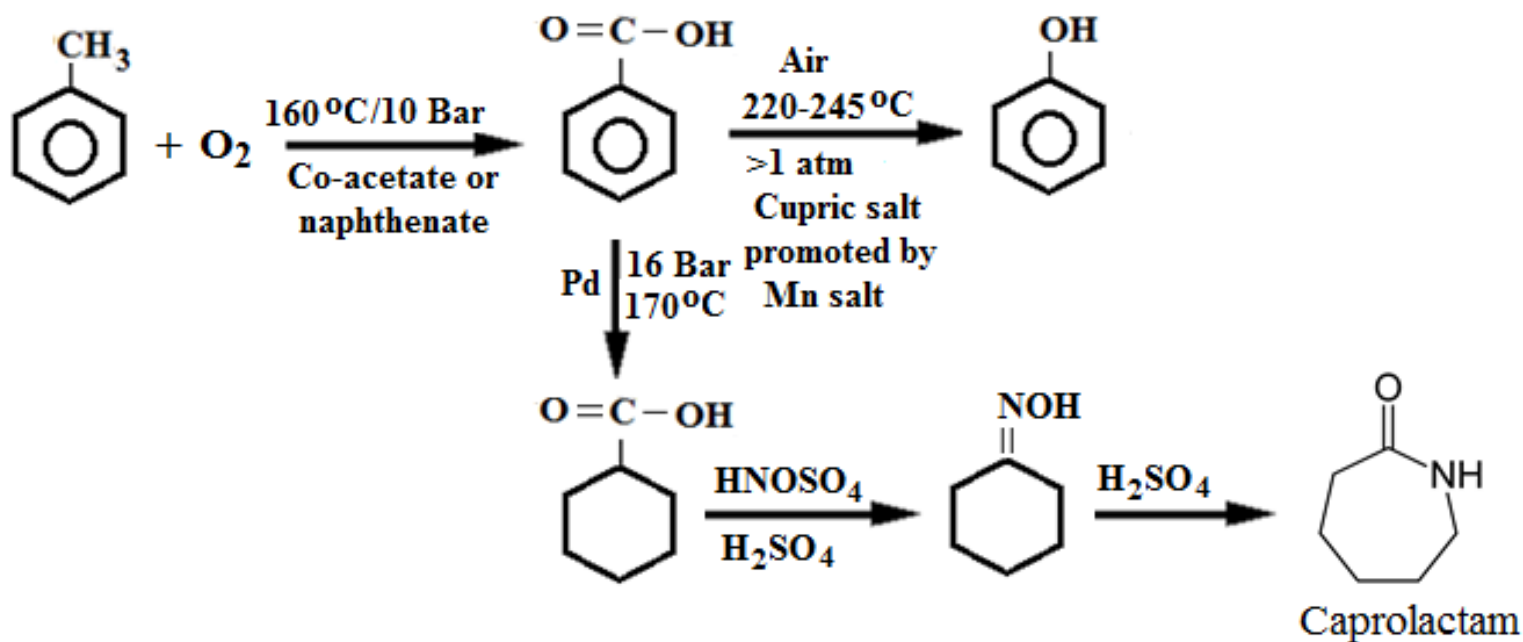
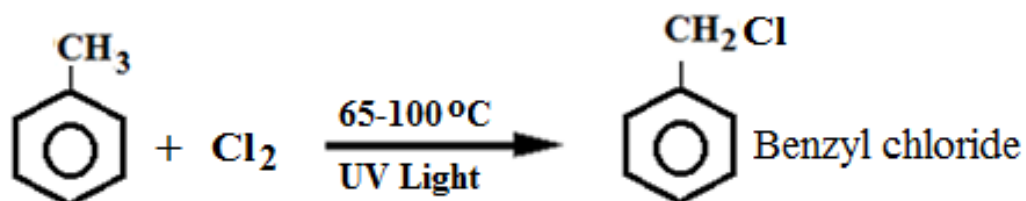
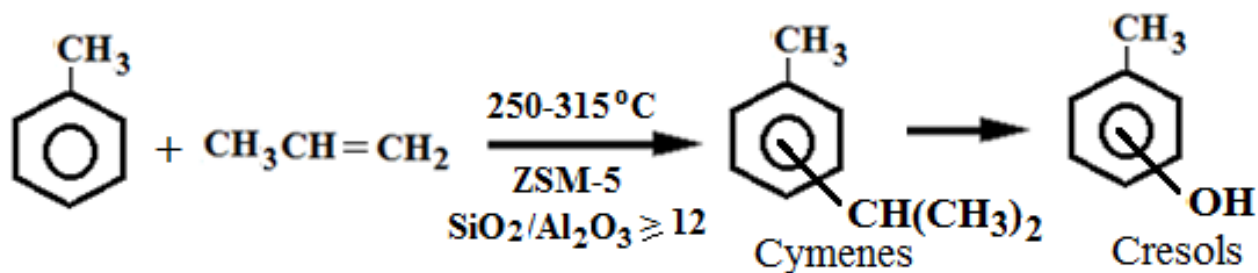
cat.  $\text{H}\ddot{\text{Cl}}\text{:}$



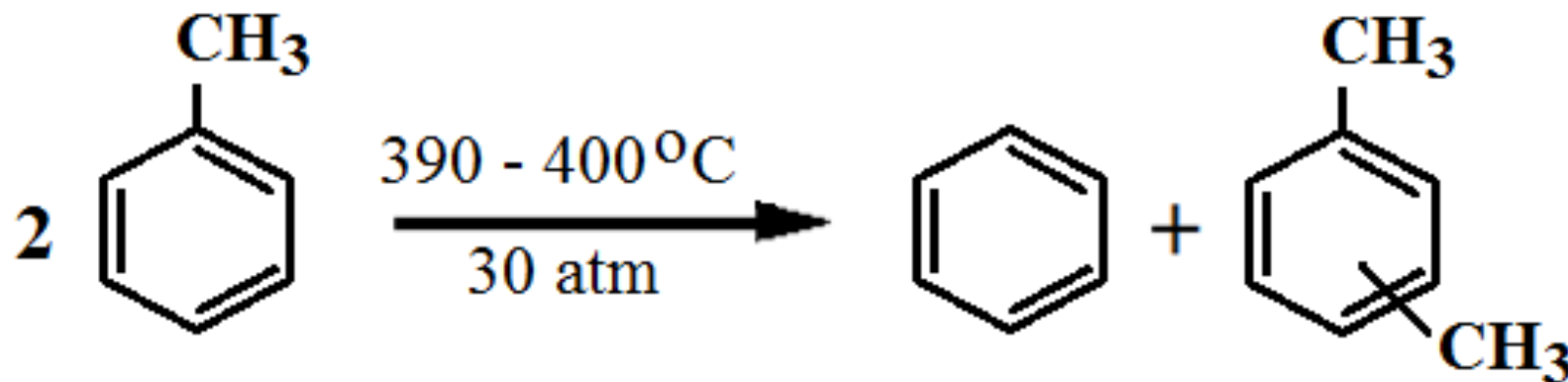
Polyurethane



## Chemicals from Toluene



# Xylenes from Toluene by Selective Disproportionation



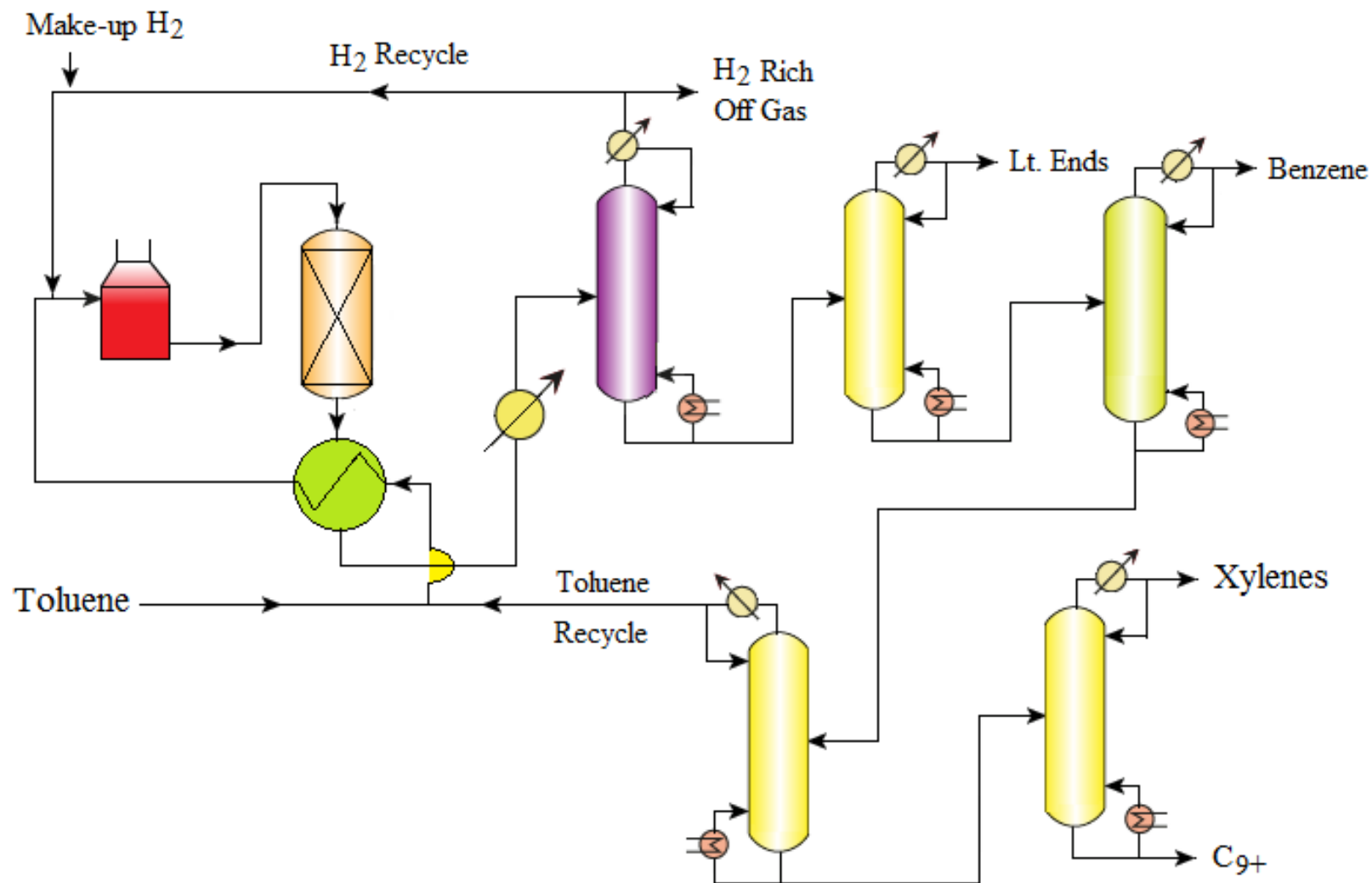
High silica Zeolite catalyst

WHSV = 4.0 - 4.5  $\text{h}^{-1}$

Toluene conversion = 28 - 30%

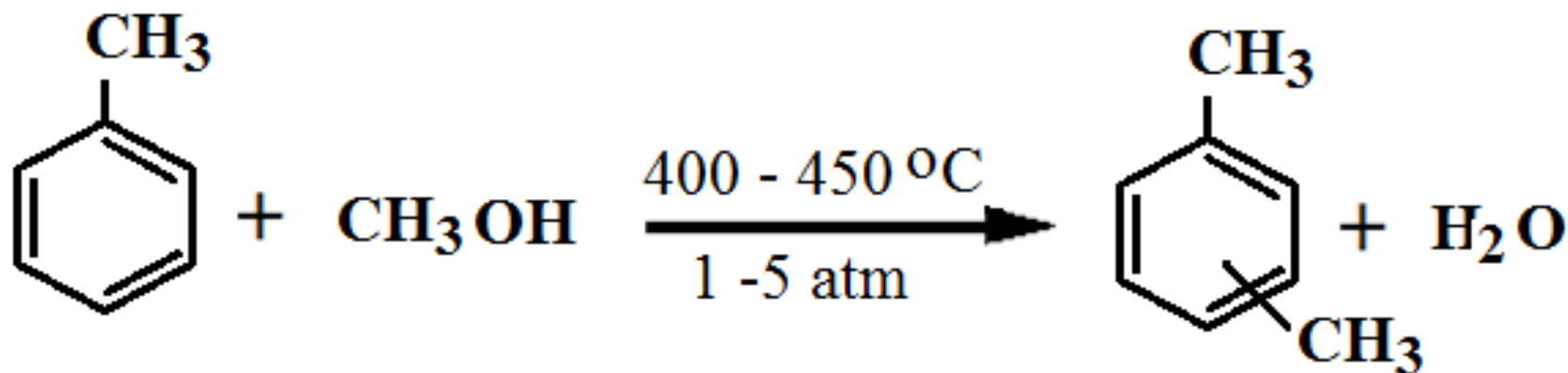
*p*-Xylene selectivity (wt%) = 80-90

**[GT-STDP Process]**



**Production of Xylenes by Selective Toluene Disproportionation (STDP)**

## Xylenes from Toluene by Alkylation with Methanol



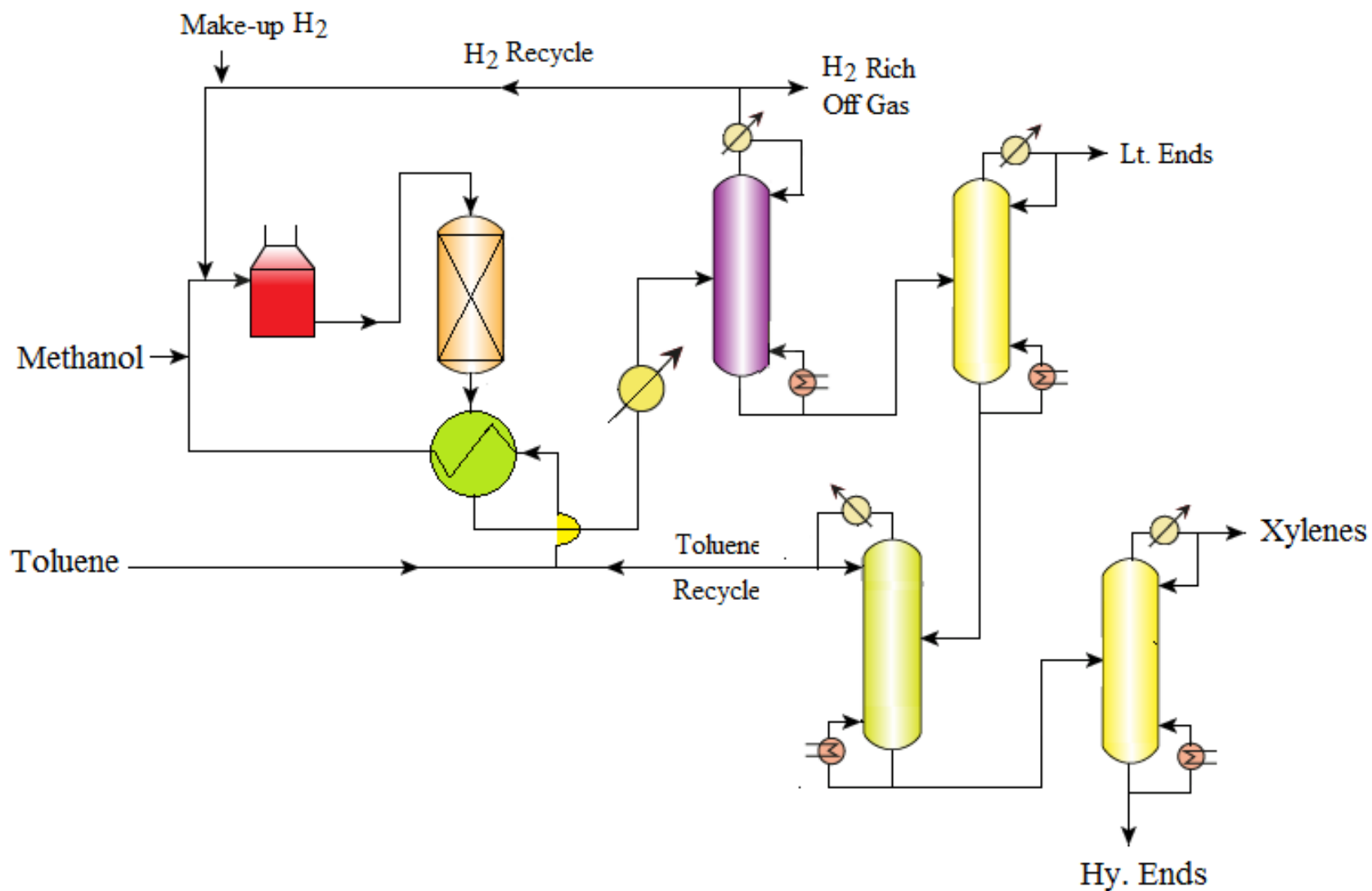
High silica Zeolite catalyst

$\text{WHSV} = 1\text{-}2\text{ h}^{-1}$

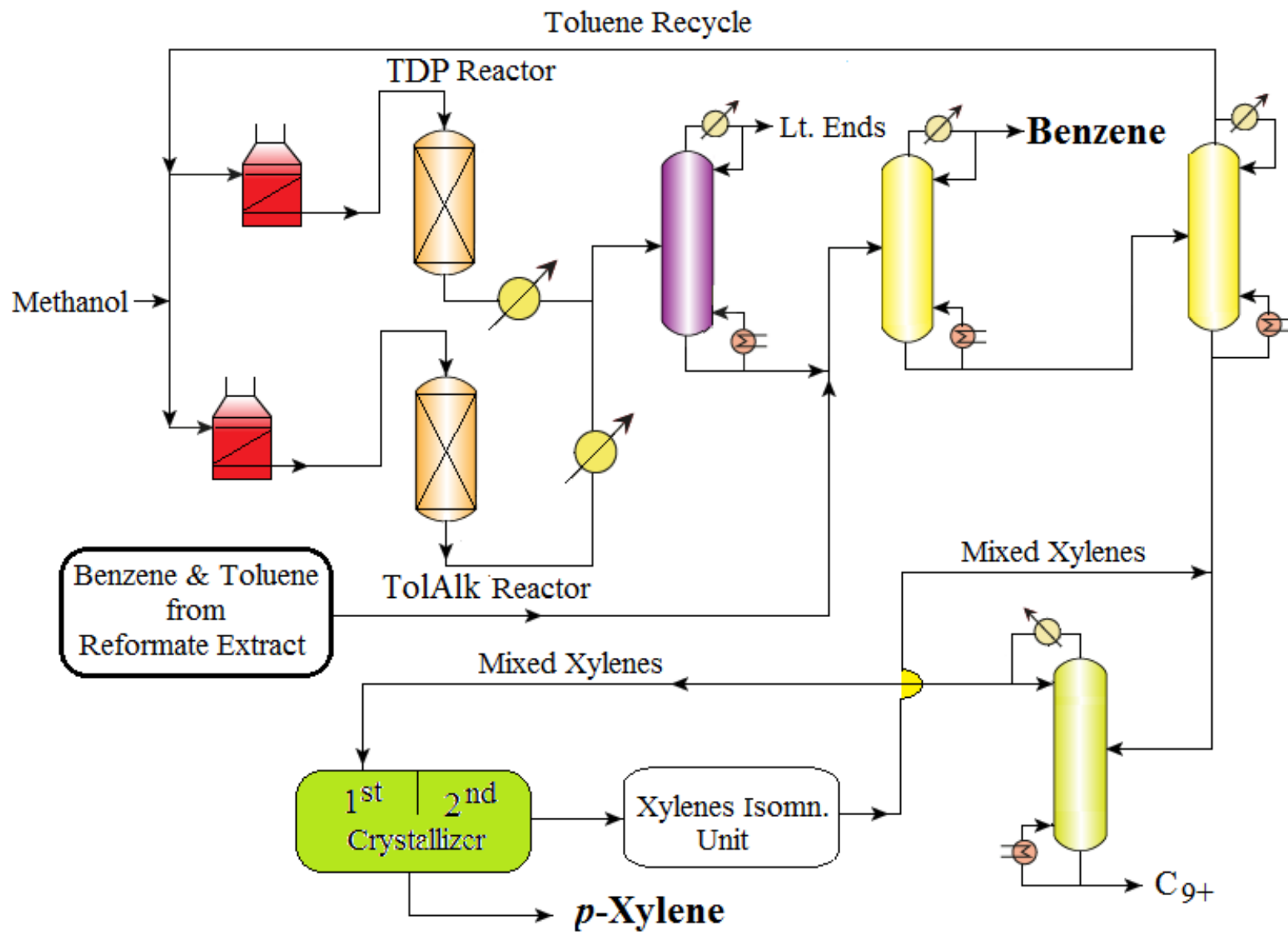
*p*-Xylene selectivity (wt%) = 80-90

**[GT-TolAlk Process]**





**Production of Xylenes by Methylation of Toluene**



**Optimized Aromatic Complex for Maximum Production of *p*-Xylene**

Thank you!